Arnob Deb
Computer Science & Engineering
Indian Institute of Technology Bombay

25M0779
M.Tech.
Gender: Male
DOB: 19/09/2002

Examination	University	Institute	Year	CPI / % Credits
Post Graduation	IIT Bombay	IIT Bombay	2027	null Core: | Total:
Graduation	MAKAUT	Haldia Institute of Technology	2024	9.42
Graduation Specialization: Computer Science & Engineering				
Intermediate	CBSE	BSF Sr Sec Residl School Kadamtala	2019	85.80%
Matriculation	CBSE	Darjeeling Public School	2017	9

RESEARCH PROJECTS & SEMINAR

- **Latency Hiding via Out-of-Order Scheduling in GP-GPUs** *(Jan'26-Apr'26)*
  *(M.Tech Seminar | Guide: **Prof. G. Sivakumar & Prof. Virendra Singh**)*
  - Investigated the **TLP ceiling** from simultaneous warp stalls and occupancy limits, surveying various **OoO scheduling** techniques via dependency tracking, pre-execution and compiler guidance for stall hiding.
  - Synthesized **intra-warp ILP** opportunities and design trade-offs, identifying **lightweight OoO**, smarter warp scheduling and memory optimizations as a promising direction for higher GPU utilization.
- **Mitigating RowHammer Vulnerability in GPU Main Memory** *(Jan'26-Apr'26)*
  *(R & D Project | Guide: **Prof. Virendra Singh** | **ISRDC Lab**)*
  - Surveyed the **RowHammer** arms race from CPU to GPU-specific exploits, isolating **GDDR6-specific** attack barriers: higher DRAM latency, shorter refresh windows, and opaque row-to-address mappings.
  - Built a **cycle-accurate** GPU memory subsystem simulator integrating **GPGPU-Sim** with **DRAMSim3** via a custom wrapper, profiling mitigation hooks — revealing the L2 cache **throttles hammering rate by 40%**.
- **Skylight Edge Server : KV-Cache Compression for Long-Context LLMs** *(Jun'26-Present)*
  *(M.Tech Project | Guide: **Prof. Aditya Desai** | **CFILT Lab**)*
  - Benchmarked **LLM inference systems** to identify **OpenVINO** as the target backend, profiling its orchestration to characterize inference flow, device dispatch, and latency across execution stages.
  - Analyzed modern **KV-cache compression** methods for long-context LLMs, comparing **memory–latency–accuracy trade-offs** to identify viable approaches for integration into the OpenVINO inference stack.
  - Implementing selected compression methods in **OpenVINO** by identifying **integration points** in the inference pipeline, targeting **reduced memory footprint**, latency and energy usage on **long-context workloads**.

KEY COURSE PROJECTS

- **Cache-Aware Matrix Multiplication and Embedding Operation Optimization** *(Aug'25-Nov'25)*
  *(CS683: Advanced Computer Architecture | Instructor: **Prof. Biswabandan Panda**)*
  - **Accelerated** naive matrix multiplication **up to 8.3×** via loop reordering, unrolling, tiling, and SIMD vectorization, then applied the optimized kernel to **speed up** neural-network forward-pass execution **by 4.3×**.
  - **Reduced** embedding-lookup latency **up to 2.92×** through tuned **software prefetching** combined with SIMD vectorization, validated using **perf**-based L1D, L2, and LLC cache-miss analysis across million-row tables.
- **Adaptive Prefetching and Cache Hierarchy Design in ChampSim Simulator** *(Aug'25-Nov'25)*
  *(CS683: Advanced Computer Architecture | Instructor: **Prof. Biswabandan Panda**)*
  - Implemented a **region-based L2 offset prefetcher** with adaptive degree control based on runtime accuracy feedback, achieving **up to 14% speedup** over baseline across varying table sizes on ChampSim traces.
  - Built an **exclusive cache hierarchy** across L1/L2/LLC in ChampSim — victim cache eviction, cross-level invalidation, quantifying **IPC** and **MPKI** trade-offs against the default non-inclusive design.
- **IITB-RISC Processor : 6-Stage Pipeline to 2-Way OoO Superscalar Architecture** *(Jan'26-Apr'26)*
  *(EE739: Processor Design | Instructor: **Prof. Virendra Singh**)*
  - Designed a **6-stage RISC pipeline** in **VHDL** with IF-ID-RR-EX-MEM-WB datapath, pipeline registers and **hazard handling** through operand forwarding and flushes minimizing unnecessary pipeline stalls.
  - Extended the pipeline to a **2-way OoO superscalar** with dual-instruction fetch, reservation stations, register renaming & dynamic scheduling, enabled parallel execution using **ROB** with in-order commit.
  - Validated the processor using a **Vivado VHDL testbench**, verifying instruction flow, dependency resolution, hazard handling and **OoO execution** across dependent and independent instruction sequences.

- **Systematic Load Testing of a Multi-Tier Employee Management Server** *(Aug'25-Nov'25)*
  *(CS744: Design and Engineering of Computing Systems | Instructor: **Prof. Mythili Vutukuru**)*
  - Built a multi-tier HTTP server with **RESTful CRUD APIs**, thread-safe 64-entry LRU caching and thread-local **MySQL** connections using **CivetWeb**, enabling concurrent request handling and persistent storage.
  - Characterized system capacity using a **closed-loop load generator** across various load levels, isolating CPU-bound vs I/O-bound bottlenecks through throughput, response time, and resource utilization.

OTHER COURSE PROJECTS

- **Modelling the Backlog of Judicial Cases for the Bombay High Court** *(Jan'26-Apr'26)*
  *(CS752: System Dynamics: Modeling & Simulation for Development | Instructor: **Prof. Om Damani**)*
  - Modeled judicial **case-backlog dynamics**, formulating a case-aging cohort chain, nonlinear burnout-driven judicial capacity, and perceived-delay feedback loops in **Vensim** calibrated to 2001–2025 pendency data.
  - Conducted **sensitivity analysis** and policy simulations on judicial appointment rates and burnout thresholds in Vensim, identifying accelerated hiring **cuts Year-30 backlog by 27%** vs an unsustainable base case.
- **Krishimitra - Empowering Farmers through Market Intelligence** *(Aug'25-Nov'25)*
  *(CS699: Software Lab | Instructor: **Prof. Om Damani**)*
  - Built a real-time **market intelligence platform** aggregating live eNAM mandi prices and historical trends to eliminate price information asymmetry, **helping farmers compare prices** and **identify profitable markets**.
  - Developed a **Smart Transport Calculator** and community forum that computes net profit across top-paying mandis nationally and state-wise, enabling farmers to make **data-driven selling decisions**.
- **Comparison of Label Smoothing vs No Smoothing in Classification** *(Aug'25-Nov'25)*
  *(CS725: Foundations of Machine Learning | Instructor: **Prof. Abir De**)*
  - Developed **CNN**-based image classification models in **PyTorch** to evaluate the impact of Label Smoothing against standard cross-entropy training on **MNIST** and **CIFAR-10** datasets.
  - Compared model performance by varying smoothing factors, demonstrating optimal smoothing **cuts CIFAR-10 ECE by 5x** while lifting test F1 from **0.76 to 0.80** and improves MNIST test accuracy from **98.6% to 99.12%**.

PUBLICATIONS

- Mazumdar, Arpita, Arnob Deb, et al. (2025). "**Employability Prediction of Engineering Undergraduates Using Machine Learning Approach**". *In: Proceedings of RAICCIT 2025: Research Advancements and Innovations in Computing, Communications, and Information Technologies. JIS University*, pp. 208–214. ISBN: 978-81-973699-3-3.

COURSES UNDERTAKEN

- CS 683: Advanced Computer Architecture
- CS 744: Design and Engineering of Computing Systems
- CS 725: Foundations of Machine Learning
- CS 631: Implementation Techniques for RDBMS *(*ongoing)*
- EE 739: Processor Design
- CS 699: Software Lab
- CS 752: System Dynamics
- CS 6013: Efficient AI *(*ongoing)*

TECHNICAL SKILLS

- **Programming Languages** : C, C++, Python, Bash, SQL, PostgreSQL, HTML, CSS, JavaScript
- **Simulation Tools**: GPGPU-Sim, DRAMSim3, ChampSim, Vivado, Vensim
- **Dev Tools**: Perf Tool, OpenVINO, PyTorch, MySQL, Git, LaTeX, Make, NumPy, Pandas
- **Core Concepts**: Computer Architecture, OS, DBMS, Computer Networks, OOPs, DSA, System Design

POSITION OF RESPONSIBILITY

- **Student Companion** | Institute Student Companion Programme (ISCP), IIT Bombay *(Jul'26-Present)*
  - Part of a **330-member team** selected via interviews, ensuring smooth transitions of new students.
  - Mentoring a **group of 5** students in academics, and extracurricular activities, ensuring their well-being.
- **Teaching Assistant** | Department of CSE, IIT Bombay
  - **CS 683 : Advanced Computer Architecture** under **Prof. Biswabandan Panda** *(Aug'26-Present)*
  - **CS 101 : Computer Programming and Utilization** under **Prof. Ashutosh Gupta** *(Jan'26-Apr'26)*
  - **CS 101 : Computer Programming and Utilization** under **Prof. Rohit Gurjar** *(Aug'25-Nov'25)*

ACHIEVEMENTS AND EXTRA CURRICULAR ACTIVITIES

- Secured **2nd position** in **Algo GC**, a Hostel General Championship event at IIT Bombay.
- Successfully completed the 5.5-kilometer Crossy GC at IIT Bombay.
- **Hobbies**: Gaming, Anime, Badminton, Pool, Football, Card Games, Building PCs, Traveling